

Name: _____

Date: _____

Unit 5 Quiz

Modern Physics

K	1	A	C
9 / 10	10 / 10	3 / 4	3 / 4

Well Done!

SHOW ALL YOUR WORK IN ANSWERING QUESTIONS.**APPLICATION**

1. A linear particle accelerator is a special machine that accelerates charged particles to nearly the speed of light and channel them into a beam. When used in research, the beam hits targets and the atoms split. Scientists gather information about sub-particles, atoms and molecules to further help understand the laws of physics.

A proton of a mass 1.67×10^{-27} kg is accelerated to 0.999 994c.

- a) What is the non-relativistic momentum in terms of c (the speed of light)?
 b) Determine the relativistic momentum in terms of c (the speed of light).
 c) Calculate the ratio of the proton's relativistic momentum to the non-relativistic momentum. (3 marks)
 Show all your work.

a) $p = mv$
 $m = 1.67 \times 10^{-27}$ kg
 $v = 0.999994c$

$p = (1.67 \times 10^{-27})(0.999994 \times 3.0 \times 10^8)$
 $p = 5.01 \times 10^{-19}$ kgm/s

b) $p = \frac{mv}{\sqrt{1 - \frac{v^2}{c^2}}}$

$m = 1.67 \times 10^{-27}$ kg
 $v = 0.999994c$
 $p = \frac{1.67 \times 10^{-27} \cdot 0.999994c}{\sqrt{1 - (0.999994)^2}}$

$p = 1.45 \times 10^{-16}$ kgm/s

c) $\frac{1.45 \times 10^{-16} \text{ kgm/s}}{5.01 \times 10^{-19} \text{ kgm/s}} = x$ [4 marks]

$x = 289.412$, relativistic is 289.412 larger than non-relativistic

Final answer: a) $p = 5.01 \times 10^{-19}$ kgm/s

Final answer: b) $p = 1.45 \times 10^{-16}$ kgm/s

Final answer: c) $x = 289.412$

COMMUNICATION

1. In a thought experiment, a person boards a spaceship on Earth and travels at 0.8c, while her twin brother remains at home on Earth. This is known as the twin paradox.

- a) Which twin is younger: the one who boards the spaceship or the one on Earth?
 b) Why do the twins age differently?
 c) How does this explain the asymmetry in the paradox?

a) The sister on the spaceship is younger, because her brother experiences time dilation & goes older. [4 marks]

b) Because this sister is travelling close to the speed of light, & the brother is not, he experiences time dilation, making more time pass for him than her.

c) According to the time dilation equation, the brother on Earth (Δt)

is much longer than the sister's time period (Δt_0).

When $v = 0.8c$, the equation looks like: $\Delta t = \frac{\Delta t_0}{\sqrt{1 - (0.8)^2}}$, $\Delta t = \frac{\Delta t_0}{0.6}$, which means the brother's Δt will always be observed as greater than the sister's Δt_0 .

THINKING

1. A spacecraft travels at $0.67c$ relative to Earth. An astronaut on board listens to a CD that lasts 50.0 min. How long would the CD last as measured by a stationary clock on Earth? Show all your work.

$$\Delta t_0 = 50 \text{ min}$$

$$v = 0.67c$$

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\Delta t = \frac{50 \text{ min}}{\sqrt{1 - 0.67^2}}$$

$$\Delta t = 67.35 \text{ min}$$

[2.5 marks]

Final answer: $\Delta t = 67.35 \text{ min}$

2. A neutron with a mass of $1.67 \times 10^{-27} \text{ kg}$ has a kinetic energy of $1.20 \times 10^{-13} \text{ J}$ as it moves through a material. Using the de Broglie wavelength formula, calculate the wavelength of the neutron in meters. Planck's constant is $6.63 \times 10^{-34} \text{ J}\cdot\text{s}$. Show all your work.

$$E_k = 1.2 \times 10^{-13} \text{ J}$$

$$m = 1.67 \times 10^{-27} \text{ kg}$$

$$E_k = \frac{1}{2}mv^2$$

$$v^2 = \frac{2E_k}{m}$$

$$v = \sqrt{\frac{2E_k}{m}}$$

$$v = \sqrt{\frac{2 \cdot 1.2 \times 10^{-13} \text{ J}}{1.67 \times 10^{-27} \text{ kg}}}$$

$$v = 1.1988 \times 10^7 \text{ m/s}$$

$$\lambda = \frac{h}{mv}$$

$$\lambda = \frac{6.63 \times 10^{-34} \text{ J}\cdot\text{s}}{1.67 \times 10^{-27} \text{ kg} \cdot 1.1988 \times 10^7 \text{ m/s}}$$

[3.5 marks]

Final answer: $\lambda = 3.31 \times 10^{-14} \text{ m}$

3. Nuclear power stations in Canada use a type of reactor invented in Canada called a Canada Deuterium Uranium (CANDU) reactor. A typical CANDU fuel bundle has a mass of 23 kg.
- a) Calculate the amount of electrical energy produced by a single CANDU fuel bundle, assuming that the entire bundle is converted to electrical energy. Show all your work.

$$E_0 = mc^2$$

$$m = 23 \text{ kg}$$

$$c = 3 \times 10^8 \text{ m/s}$$

[2 marks]

$$E_0 = 23 \text{ kg} \cdot (3 \times 10^8 \text{ m/s})^2$$

$$E_0 = 2.07 \times 10^{18} \text{ J}$$

Final answer: $E_0 = 2.07 \times 10^{18} \text{ J}$

- b) An average Canadian home uses $3.6 \times 10^{10} \text{ J}$ of energy per day. Determine the number of days that a single CANDU fuel bundle could theoretically provide energy to a home. Show all your work.

$$\text{Avg energy/day} = 3.6 \times 10^{10} \text{ J/day}$$

$$E_0 = 2.07 \times 10^{18} \text{ J}$$

$$t = \frac{E_0}{\text{Avg energy}}$$

$$t = \frac{2.07 \times 10^{18} \text{ J}}{3.6 \times 10^{10} \text{ J/day}}$$

$$t = 5.75 \times 10^7 \text{ days}$$

[2 marks]

Final answer: $5.75 \times 10^7 \text{ days}$